# A formal knowledge graph for integrating metagenomics and geochemistry of the Rub' al-Khali desert using the Semanticscience Integrated Ontology

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**Abstract**

Integrating multi-omic data with environmental metadata requires a unified data model that can harmonize heterogeneous observation types, enforce structural constraints at ingest, support logical consistency checking, and provide semantic query answering at scale. We present a formal knowledge graph that integrates 16S rRNA gene amplicon profiles, X-ray fluorescence measurements, field observations, and climate records from a five-expedition sampling campaign across the Rub' al-Khali desert (the companion data paper reports the expedition and data acquisition in detail). The knowledge graph uses the Semanticscience Integrated Ontology (SIO) as the upper-level framework, with 302 domain classes grounded in ChEBI, ENVO, PATO, the Units Ontology, and NCBI Taxonomy. We formalize eight reusable ontology design patterns—for measurement, sampling, experimental protocol, taxonomic abundance, X-ray fluorescence geochemistry, sampling site with biome assignment, the sequencing workflow, and sequencing quality control—in description logic and align them explicitly with SIO's Process–hasInput/hasOutput–Object, Collection–hasMember–Thing, and Quantity–refersTo–Quality idioms. A Shape Expressions (ShEx) validation step enforces structural shapes on every RDF batch before it is loaded into the triple store, and the ELK reasoner confirms logical consistency of the full knowledge graph (zero unsatisfiable classes) both before and after materialization. An OWL-Horst (pD*) forward-chaining materialization workflow in Virtuoso physically expands the graph from 37.0 million asserted triples to 117.9 million total triples (a 3.18× expansion) in approximately 114 minutes. We evaluate the deployed resource through nine SPARQL competency questions with sub-second to second-range latencies, and we serve the knowledge graph through a public SPARQL 1.1 endpoint and a web portal with an autocomplete-aware query editor. We discuss the design rationale for every major choice, connect each choice to the underlying challenge, and identify limitations and directions for future work.

**Keywords.** knowledge graph; SIO; OWL; ontology design patterns; OWL-Horst; materialization; ELK reasoner; ShEx; Virtuoso; SPARQL; desert microbiome; Rub' al-Khali

## 1 Introduction

Characterizing a complex ecosystem requires integrating heterogeneous multi-scale data: microbial community profiles derived from DNA sequencing, elemental soil composition from geochemical analyses, spatiotemporal environmental variables, and provenance metadata that links physical samples to their digital representations [14]. Traditional approaches store these data types in separate, disconnected repositories, which limits cross-domain queries and reduces reproducibility. The emerging concept of an ecosystem *digital twin* [3, 22, 23, 25] depends on precisely the kind of unified, machine-interpretable, interoperable data model that is missing for most natural environments; while biological systems and specific regions have semantic digital representations [31], no such resource exists for the Rub' al-Khali, the largest contiguous sand desert on Earth [30], whose microbial communities and biogeochemistry remain poorly characterized [29].

A companion data paper describes the underlying dataset: a five-expedition soil sampling campaign (2023–2025) across a 1,043 km transect of the Rub' al-Khali, yielding 16S rRNA gene amplicon profiles from 2,535 soil samples at 70 sampling sites, X-ray fluorescence measurements for 91 chemical analytes from 249 measurement sessions, GPS-tagged environmental measurements recorded at every site visit, and monthly and annual climate records retrieved from the Open-Meteo database [12, 34]. This paper describes the *semantic layer*: the ontology engineering, the modular OWL/RDF architecture, the reasoning workflow, and the evaluation of the deployed resource as a Semantic Web application.

Constructing such a knowledge graph for an entire desert ecosystem requires addressing several interlocking challenges that have shaped the design of this work. First, the data are produced by heterogeneous instruments and protocols (16S amplicon sequencing, X-ray fluorescence, GPS-tagged field notes, gridded climate reanalysis), so a *unified data model* is needed that can represent samples, processes, qualities, and quantities under a common upper-level vocabulary rather than as disconnected tables. Second, taxonomic labels are reported by multiple primer choices and classifier workflows and must be *harmonized* against a single reference taxonomy without losing the original assertions. Third, because data are deposited in batches over five field expeditions, *schema validation at ingest* is required to catch structural errors before they propagate into the graph. Fourth, the knowledge graph must support *semantic query answering at scale*: with tens of millions of triples, expressive OWL 2 DL reasoning is computationally infeasible, so a tractable reasoning regime must be selected and its trade-offs made explicit. Fifth, a *user-facing query and exploration interface* is required so that domain scientists who are not fluent in SPARQL can interact with the resource. We return to each of these challenges in Section 6 and link them to the corresponding design choices in the later sections.

This paper makes the following contributions to the Semantic Web community. We present an OWL 2 DL knowledge graph that integrates a desert ecosystem dataset under the Semanticscience Integrated Ontology (SIO) [8] as the upper-level framework, with domain classes grounded in ChEBI [7], ENVO [4], PATO [15], the Units Ontology [16], and NCBI Taxonomy [41]. We formalize eight reusable ontology design patterns in description logic and align them explicitly with SIO's core relational idioms, accompanied by Turtle listings and prose verbalizations so that readers can reproduce and extend them. We integrate Shape Expressions (ShEx) [37] validation into an automated deployment workflow as a regression test against structural drift. We implement OWL-Horst (pD*) [43] forward-chaining materialization as three rounds of SPARQL Update statements executed inside Virtuoso [35], producing a 3.18× expansion from 37 million asserted to 118 million total triples in approximately 114 minutes. We validate the logical consistency of the full graph with the ELK reasoner [24] both before and after materialization and show that the materialization introduces no disjointness violations. We report compute-time metrics for the key deployment steps and per-query latencies for a suite of nine SPARQL competency questions, and we serve the resource through a public SPARQL 1.1 endpoint and a web portal with an autocomplete-aware query editor. Finally, we discuss the design rationale for every major choice, the current limitations, and a sustainability plan for future data increments and upstream ontology updates.

## 2 Related work

The Semantic Web community has produced a rich ecosystem of standards, vocabularies, and tools that collectively cover parts of the task of describing an ecological dataset. This section first surveys those standards, then reviews the reasoning and deployment literature that our work builds on, and finally articulates the gap that motivates our design: no single existing standard covers the full provenance chain from a physical soil sample to an integrated, materialized knowledge graph that supports cross-domain query answering, and combining them requires an integrative layer of the kind we provide.

### 2.1 Upper-level ontologies for scientific data integration

The Semanticscience Integrated Ontology (SIO) [8] is a lightweight upper-level ontology that provides general-purpose classes and relations applicable across scientific domains, including entities (objects, processes, attributes), qualities, quantities, and collections. SIO has been used to describe biomedical datasets, laboratory experiments, and observational studies, and it is compatible with several community vocabularies (OBO Foundry, Units Ontology, ChEBI). We adopt SIO because its relational idioms (Process–hasInput/hasOutput–Object, Collection–hasMember–Thing, and Quantity–refersTo–Quality) generalize uniformly across the sampling, laboratory, and computational processes that our dataset requires. Recent work on simplified upper-level ontologies, notably SULO [9], takes this argument further: by strengthening a few fundamental relations and reducing reliance on specialized predicates, an upper-level ontology can lower the modelling effort for new domains while preserving interoperability. Our design patterns are compatible with this direction and could be re-grounded in SULO with limited refactoring.

### 2.2 Related standards for provenance, observations, and sampling metadata

Several Semantic Web standards address subsets of the problem we solve. We briefly review the most relevant, grouped by concern, and explain in each case what the standard covers and what it leaves uncovered.

**Provenance: PROV-O.** The W3C PROV-O [27] recommendation defines a small number of core classes (`Activity`, `Entity`, `Agent`) and relations (`used`, `wasGeneratedBy`, `wasDerivedFrom`, `wasAssociatedWith`) that are sufficient to record *that* one data item was derived from another, by whom, and when. PROV-O is deliberately domain-agnostic and does not model what the activity *does* beyond "used input $X$, produced output $Y$." In our setting, every scientific process—sampling, DNA extraction, library preparation, sequencing, taxonomic classification, XRF analysis—is structurally similar at the PROV-O level but semantically distinct: a sampling process targets a site, an XRF process targets a sample and produces typed concentration quantities, a bioinformatic workflow produces a taxonomic dataset whose members are abundance observations. Encoding these domain semantics in pure PROV-O forces the modeller to introduce custom sub-classes of `Activity` and `Entity` for every distinct process and data type. Our SIO-based patterns give this structure directly in a single upper-level framework, and every instance can be projected onto PROV-O as needed because SIO's Process–hasInput/hasOutput–Object relations map one-to-one onto PROV-O.

**Observations and sensors: SOSA/SSN.** The W3C/OGC SOSA (Sensor, Observation, Sample, Actuator) vocabulary and the Semantic Sensor Network (SSN) ontology [19] define classes for `Observation`, `Sensor`, `Sample`, `FeatureOfInterest`, and `ObservableProperty`. SOSA/SSN is oriented primarily toward Internet-of-Things deployments, continuous sensor streams, and the relationship between a physical sensor and the property it measures. It is well-suited for settings where the measurement process is a thin wrapper around a sensor reading. Our dataset does not fit that mould: we have one-off field measurements made with handheld instruments, batch laboratory workflows, and computational analyses whose "observable property" (for example, the relative abundance of a taxon in a 16S rRNA dataset) is itself the output of a multi-step bioinformatics workflow. SOSA's `Sample` class is intentionally thin and does not capture the soil-compartment / replicate structure of our sampling design. Furthermore, SOSA does not model the provenance chain from sample to digital data product. Our design patterns could in principle be rephrased over SOSA by treating each laboratory and computational step as a `hosts` relation on a virtual sensor, but this mapping is unnatural and loses the process-centric semantics that make the ODPs re-usable across domains.

**Experimental protocols: OBI.** The Ontology for Biomedical Investigations (OBI) [2] provides a rich vocabulary for laboratory workflows, protocols, agents, devices, and experimental designs in the biomedical domain. OBI's coverage includes many of the process patterns we need—material processing, library preparation, sequencing assays—and its alignment with BFO and IAO gives it a sound upper-level grounding. Two considerations kept us from adopting OBI as our sole upper-level framework. First, OBI is heavily oriented toward clinical and biomedical investigations; many of its classes assume a human subject or a clinical assay context that does not apply to environmental sampling. Second, OBI is large and has a steep onboarding cost for a focused dataset like ours. We therefore use OBI indirectly through SIO compatibility: several of our DNA extraction and sequencing classes could be re-aligned with OBI classes as a future extension, and we already import ChEBI and PATO which are in the OBO family OBI uses.

**Minimum metadata for sequencing: MIxS.** The Genomic Standards Consortium's MIxS checklist [46] specifies the minimum metadata fields required for sequence submissions (MIMARKS for marker genes, MIGS/MIMS for genomes and metagenomes). MIxS is a flat key-value checklist rather than an ontology: it prescribes *which* fields must be reported and what controlled vocabularies to draw values from, but provides no formal semantics and no inference. It is sufficient for ENA submission but insufficient for cross-domain reasoning. A MIxS record can be projected into our knowledge graph as a set of triples about a sample, an environmental context, and an associated sequencing run, but the inverse projection from our knowledge graph to a MIxS record is trivial: every field the checklist requires corresponds to a SPARQL query against an instance of our sampling and sequencing patterns. In that sense we treat MIxS as a downstream consumer of our knowledge graph rather than an upstream data model.

**Ecological observations: OBOE.** The Extensible Observation Ontology (OBOE) [28] provides a general structure for ecological observations based on four core classes: `Observation`, `Entity`, `Measurement`, and `Characteristic`. OBOE has influenced the design of SOSA/SSN and is conceptually close to the measurement pattern we use. Two reasons kept us from adopting OBOE directly: its community uptake has remained modest compared with SIO and SOSA, and the observation-centric framing does not extend naturally to the laboratory and computational processes that dominate our provenance chain. Where an observation in OBOE is a single event, our pipeline has many events—sampling, extraction, amplification, sequencing, bioinformatic analysis—each of which generates its own quantities and qualities. Modelling this as a chain of OBOE observations is possible but loses the PROCESS–HASINPUT/HASOUTPUT–OBJECT abstraction that makes the patterns composable.

**Biodiversity records: Darwin Core.** Darwin Core [44] is the dominant community standard for biodiversity occurrence records and underlies the Global Biodiversity Information Facility (GBIF). Darwin Core is primarily a flat record format with a controlled vocabulary (Darwin Core Terms) rather than a semantic ontology, though a Darwin-Core-in-RDF mapping exists. The scope of Darwin Core is the occurrence record—"this organism was observed at this place at this time"—and it does not model laboratory or computational processes. It can be linked in as a complementary vocabulary for the taxonomic component of our knowledge graph, and our per-sample taxon observations could be projected into Darwin Core records for GBIF deposit, but Darwin Core alone does not provide the integrative data model we need.

**Units of measurement: QUDT and OM.** Both QUDT [38] and OM [40] provide comprehensive RDF-based vocabularies for units of measurement, quantities, and dimensions. They are strong alternatives to the Units Ontology (UO) [16]. We use UO because it is part of the OBO Foundry and therefore integrates cleanly with ChEBI and PATO in a single import closure.

QUDT and OM could be swapped in without changing any of our ontology design patterns; the measurement pattern (Equation 1) is unit-vocabulary-agnostic.

**Information artefacts: IAO.** The Information Artifact Ontology (IAO) [6] formalizes information content entities, documents, data items, and measurement data. IAO is widely used in OBO-family resources for distinguishing the digital artefact that records a measurement from the measurement process itself. Our knowledge graph makes this distinction implicitly through the SIO `Quantity` class and the data-property links back to literal values; a future alignment of our quantity classes with IAO would make the distinction explicit at the cost of an additional import.

### 2.3 What this work adds

The preceding survey shows that each existing standard covers a proper subset of the integration task. PROV-O provides provenance relations but not process semantics. SOSA/SSN provides observations but is oriented toward sensor streams and thin samples. OBI provides biomedical laboratory workflows but is heavy and biomedical-focused. MIxS provides a reporting checklist but no formal semantics. OBOE provides ecological observations but has limited uptake and observation-centric framing. Darwin Core provides biodiversity records but no laboratory or computational provenance. None of these standards alone provides a unified data model that can represent sampling sites, field observations, laboratory processes, sequencing workflows, bioinformatic analyses, and taxonomic abundance observations under a common vocabulary, with full provenance, with in-workflow structural validation, and with logical consistency checking at deployment time. Four specific gaps drive our work:

- **Heterogeneity of process types.** The ETL for a field expedition touches at least six distinct process categories: field sampling, field measurement, laboratory extraction, laboratory measurement, sequencing, and computational analysis. Each category is served by a different community standard in the literature (OBOE/SOSA for field, OBI for laboratory, MIxS for sequencing, PROV-O or OBOE for analysis). Combining them requires either a Frankenstein import (every standard contributes its own upper-level classes) or a thin integrative layer. We chose the latter: a single SIO-based upper layer under which each process category instantiates the same PROCESS–HASINPUT/HASOUTPUT–OBJECT idiom. Our contribution is the specific set of eight design patterns (Section 4.3) that show how this layering works in practice, with formal DL and executable Turtle for every pattern.
- **Full provenance chain with uniform IRIs.** Existing resources that cover parts of this chain typically use different identifier schemes at different stages: the physical sample is identified by an institutional barcode, the DNA extract by a laboratory accession, the sequencing run by an ENA accession, the taxonomic classification by an OTU string, and the analysis output by a file path. Cross-referencing these requires manual joining. Our knowledge graph assigns every entity—site, visit, sample, DNA extract, amplicon library, sequencing run, FASTQ dataset, taxonomic observation, QC measurement—a single persistent IRI under the `https://rubalkhali.science/kb/` namespace, and connects them through SIO's `hasInput`/`hasOutput` relations. As a result, a single SPARQL traversal can walk from a taxon abundance back to the soil sample and on to the sampling site and the environmental measurements recorded at that visit, as demonstrated by Listing 12. No existing standard provides this provenance chain out of the box; it has to be built by the integrator. We provide it.
- **Reasoning and validation as deployment steps.** None of the standards reviewed above specify how to validate data at ingest time or how to materialize entailments at deployment time. Community practice is to treat validation and reasoning as separate,

optional, downstream tools. We make both first-class steps in the deployment workflow: every ABox module is validated against a ShEx shape before it reaches Virtuoso, and the deployed graph is materialized via an explicit OWL-Horst forward-chaining procedure that is scripted as SPARQL Update statements inside the triple store. The materialization choice reflects a specific engineering trade-off—ELK is used for consistency checking (sound and complete for OWL 2 EL classification), while OWL-Horst is used for forward-chaining ABox expansion (a different, incomplete, but practically sufficient semantics)—that existing standards do not prescribe. Our contribution is the concrete recipe (Section 4.4) showing how to combine these two reasoners in a real Virtuoso deployment.

- **A reusable ontology design pattern library with verbalized listings.** Our eight design patterns are designed to be reused. Each is formalized in description logic, illustrated with a concrete Turtle listing, verbalized in prose, and documented in the deposited ontology with natural-language annotations. This combination—formal, instance-level, and prose—is uncommon in practice: community ontologies typically provide either formal axioms without examples or example files without formal patterns. The former is opaque to domain scientists; the latter is under-specified. We provide both, and we release the patterns as a standalone contribution so that authors of future environmental knowledge graphs can adopt them directly.

In short, SIO and the standards reviewed above give us the building blocks: SIO for the upper-level relations, ChEBI and ENVO and NCBI Taxonomy for the grounded domain terms, PROV-O for optional projection, MIxS for optional downstream export. What we add is the explicit, documented, reusable *integration layer* that binds them together for a real-world environmental dataset and runs as a validated, materialized, queryable deployment.

### 2.4 Large-scale OWL reasoning

Reasoning over the combination of an expressive TBox and a large ABox has been a recurring challenge in the Semantic Web community for two decades. HermiT [17], Pellet, and other full OWL 2 DL reasoners implement tableau algorithms that are sound and complete for the full profile but do not scale to the kind of 37-million-triple ABox we have here on commodity hardware. Three broad strategies have been developed to address scale. The first is *materialization* of entailments under a restricted semantics: ter Horst's pD* / OWL-Horst rules [43] define a set of RDF inference rules that cover most of RDFS, subproperties, inverses, and transitivity, and can be applied forward-chaining to populate an inferred graph. Virtuoso implements OWL-Horst as a backward-chaining rule set that is activated at query time, but also supports forward-chaining materialization via SPARQL Update. The second strategy is *query rewriting* under the OWL 2 QL profile, which allows the reasoner to rewrite the user's query to account for the TBox without materializing any new triples. The third strategy is *consequence-based reasoning* for restricted profiles such as OWL 2 EL; the ELK reasoner [24] is a polynomial-time consequence-based classifier for OWL 2 EL that scales to very large TBoxes with millions of named classes. In this work we combine two of these strategies: we use ELK for logical consistency checking and classification of the TBox (where its polynomial guarantees make the check tractable), and we use OWL-Horst forward-chaining materialization in Virtuoso to expand the ABox so that query answering against the deployed endpoint can be served from physical triples without paying a per-query reasoning cost. We report quantitative results for both components below.

### 2.5 Triple store deployment

Several triple stores provide inference support and SPARQL query answering at production scale. We deployed Virtuoso [35] because it is available as free software under the GNU General

Public License, supports OWL-Horst forward-chaining materialization, and has demonstrated operational maturity in large-scale linked-data deployments.

### 2.6 Structural validation

Shape Expressions (ShEx) [37] and the Shapes Constraint Language (SHACL) are the two main W3C-compatible languages for describing and validating RDF graph structures. We chose ShEx because the data-generation workflow is batch-oriented and can invoke the validator from the same Groovy/Jena toolchain that produces the RDF. SHACL is an obvious alternative and is supported in-database by several triple stores; a future port of the shapes from ShEx to SHACL would enable in-database validation at load time. We return to this point in the discussion.

### 2.7 Ecological and environmental knowledge graphs

Several groups have published linked-data resources for biodiversity, biogeography, and environmental data (for example GBIF, GloBI, and various OBI-aligned ecosystem studies). Compared with these resources, our work differs in two respects. First, it targets a single geographic region and a single coordinated sampling effort, allowing the knowledge graph to encode the full provenance chain from physical sample to digital data product with uniform IRIs throughout. Second, the dataset tightly couples microbiome sequencing with soil chemistry and environmental measurements, which is unusual at this scale and supports cross-domain queries that are difficult or impossible to express against the individual source repositories.

## 3 Dataset overview

The companion data paper [12] describes the expedition, sampling protocol, DNA extraction, sequencing workflow, taxonomy harmonization, and environmental data retrieval in detail. We summarize the relevant figures here so that this paper remains self-contained.

The dataset is the product of five field expeditions conducted between early 2023 and late 2025 along a 1,043 km transect of the Rub' al-Khali from the western escarpment (19.15°N, 45.17°E) to the eastern border with Oman (20.71°N, 54.73°E). At each of 70 sampling sites we collected bulk surface, bulk subsurface ("deep"), and rhizosphere soil samples in three biological replicates, for a total of 2,535 samples. We generated 16S rRNA amplicon sequencing profiles for every sample on Illumina MiSeq, processed the reads with nf-core/Ampliseq + DADA2 [5,13,42], classified ASVs against the Genome Taxonomy Database (GTDB) [36], and harmonized the resulting classifications across primer/classifier combinations against NCBI Taxonomy [41] and the iNaturalist taxonomy, yielding 1,401,008 taxon–sample abundance observations. For a subset of the sites we generated X-ray fluorescence measurements for 91 chemical analytes in 249 measurement sessions, and at every site visit we recorded temperature, atmospheric pressure, and relative humidity with handheld field instruments. We additionally retrieved historical climate data from the Open-Meteo database for every site, yielding 12,936 monthly and 582 annual climate records.

We distribute the integrated knowledge graph as nine modular OWL/RDF files listed in Table 1. The modules together encode the full provenance chain from a physical sample to every digital data product derived from it.

## 4 Knowledge representation

We developed a formal knowledge base to integrate the different data types generated by the project. The semantic layer encodes the full provenance chain from physical sampling to digital

Table 1: Distributed knowledge graph modules.

| File | Contents |
|---|---|
| `rubalkhali.owl` | Base ontology (TBox); 302 domain classes grounded in SIO, ChEBI, ENVO, PATO, UO. |
| `rubalkhali_sites.owl` | 70 sampling sites with coordinates and ENVO biome annotations. |
| `rubalkhali_samples.owl` | 2,535 soil samples with compartment type and replicate identifier. |
| `rubalkhali_measurements.owl` | Environmental measurements and 12,936 monthly / 582 annual climate records. |
| `rubalkhali_xrf.owl` | 6,674 XRF measurement values across 91 analytes from 249 sessions. |
| `rubalkhali_dna.owl` | 2,410 DNA extracts with concentration measurements, agents, kits, protocols. |
| `rubalkhali_sra.owl` | 1,242 sequencing run records linked to ENA accessions. |
| `rubalkhali_qc.owl` | Sequencing quality metrics (GC content, duplicate rate, sequence counts). |
| `rubalkhali_taxonomy_abox.owl` | 1,401,008 taxon–sample abundance observations. |

data products, enabling cross-domain queries and ensuring data interoperability through shared vocabularies.

### 4.1 Architecture

The knowledge base is constructed in description logic [1] using a modular architecture that strictly separates the terminology (Terminological Box, TBox, or ontology) from the observed facts (Assertion Box, ABox, or domain knowledge). The TBox defines the ontology of the knowledge graph using OWL 2 DL [33], while the ABox contains the instance data generated from the expedition. We deployed the complete knowledge base across three named RDF graphs: an ontology graph (17,025 triples) containing the TBox, a knowledge base graph containing the asserted instance data, and a taxonomy reference graph containing the aligned taxonomy module.

### 4.2 Prefix conventions

For brevity, the Turtle and SPARQL listings throughout this paper use the prefix declarations of Table 2; these declarations should be prepended to any listing that the reader wishes to execute against the public endpoint. With these declarations in place, identifiers such as `https://rubalkhali.science/kb/RAK_P100001` are abbreviated to `rak:P100001` and `http://semanticscienc` to `sio:000291` without loss of information.

### 4.3 Ontology engineering and design patterns

We use the Semanticscience Integrated Ontology (SIO) [8] as the upper-level framework. SIO provides general-purpose classes and relations applicable across scientific domains, ensuring semantic interoperability with other SIO-based datasets. We defined all domain-specific classes as subclasses of SIO classes, and all introduced relations are sub-relations of SIO relations. We mapped classes to SIO and maintained full semantic mappings in our public repository (Section 9).

The core ontology comprises 302 distinct domain-specific classes integrated directly under the SIO framework. To accurately represent the complex relationships between environmental samples, laboratory processes, and bioinformatic workflows, we incorporated 4 object properties and 18 datatype properties. In total, the terminology model encapsulates 1,290 formal axioms.

Table 2: Prefix declarations used in all Turtle and SPARQL listings.

| **Prefix** | **Namespace** |
|---|---|
| `rak:` | `https://rubalkhali.science/kb/RAK_` |
| `sio:` | `http://semanticscience.org/resource/SIO_` |
| `pato:` | `http://purl.obolibrary.org/obo/PATO_` |
| `uo:` | `http://purl.obolibrary.org/obo/UO_` |
| `envo:` | `http://purl.obolibrary.org/obo/ENVO_` |
| `ncbitaxon:` | `http://purl.obolibrary.org/obo/NCBITaxon_` |
| `obo:` | `http://purl.obolibrary.org/obo/` |
| `rdf:` | `http://www.w3.org/1999/02/22-rdf-syntax-ns#` |
| `rdfs:` | `http://www.w3.org/2000/01/rdf-schema#` |
| `xsd:` | `http://www.w3.org/2001/XMLSchema#` |
| `geo:` | `http://www.opengis.net/ont/geosparql#` |
| `dcterms:` | `http://purl.org/dc/terms/` |

This compact semantic structure maintains precise alignment with our standardized schemas while minimizing redundant inferences.

The following eight subsections present the ontology design patterns used in the knowledge graph. Each pattern is formalized in description logic, illustrated with a Turtle listing, and verbalized in prose so that readers unfamiliar with OWL or Turtle can follow the example. All patterns explicitly instantiate SIO's core relational idioms (Process–hasInput/hasOutput–Object, Collection–hasMember–Thing, Quantity–refersTo–Quality), and later patterns reuse the earlier ones as building blocks.

### 4.3.1 Measurement pattern

We model a quantitative measurement using SIO's ontology design pattern for measurement processes: a measurement is a process that generates a value for a specific quality of a target entity.

$$\text{MeasurementProcess} \sqsubseteq \text{Process} \tag{1}$$

$$\text{Quality} \sqsubseteq \text{Attribute} \tag{2}$$

$$\text{Quantity} \sqsubseteq \text{InformationContentEntity} \tag{3}$$

$$\text{MeasurementProcess} \sqsubseteq \exists\text{hasTarget.Entity} \sqcap \exists\text{hasOutput.Quantity} \tag{4}$$

$$\text{Quantity} \sqsubseteq \exists\text{hasValue.xsd:double} \sqcap \exists\text{hasUnit.Unit} \tag{5}$$

We identify all units with classes from the Units Ontology (UO) [16]. Qualities are expressed using the Phenotype and Trait Ontology (PATO) [15] and linked to SIO as subclasses of the *Quality* class. Listing 1 illustrates an environmental temperature measurement taken at a sampling site.

Listing 1: Turtle representation of an environmental temperature measurement.

```
rak:P000001 a rak:0000006 ; # Measurement Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:4000001 . # hasOutput (Value)

rak:4000001 a rak:0000010 ; # Quantity
  sio:000221 uo:0000027 ; # hasUnit (degree Celsius)
  rak:2000003 "20.7"^^xsd:double ; # hasValue
  sio:000215 rak:5000001 . # refersTo (Quality)

rak:5000001 a pato:0000146 ; # Temperature Quality
  sio:000011 rak:1000001 . # isAttributeOf
```

In words, an instance `rak:P000001` of a measurement process has as its target the sampling site `rak:1000001` and produces as output the quantity `rak:4000001`. That quantity carries the literal value `20.7`, is reported in degrees Celsius (`uo:0000027`), and refers to a temperature quality `rak:5000001` which is in turn an attribute of the same site. Reading the chain in either direction therefore connects the numeric reading to both the measured site and the abstract quality that it quantifies.

### 4.3.2 Sampling pattern

A sampling event is a process that derives a sample from a feature of interest (the site).

$$\text{SamplingProcess} \sqsubseteq \text{Process} \tag{6}$$
$$\text{Sample} \sqsubseteq \text{MaterialEntity} \tag{7}$$
$$\text{SamplingProcess} \sqsubseteq \exists\text{hasTarget.Site} \sqcap \exists\text{hasOutput.Sample} \tag{8}$$
$$\text{Sample} \sqsubseteq \exists\text{isDerivedFrom.Site} \tag{9}$$

Listing 2 illustrates the semantic representation of a surface soil sample collected during Trip 1.

Listing 2: Turtle representation of a sampling process.

```
rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:6000001 a rak:0000021 ; # Deep Soil Sample
  sio:000252 rak:1000001 . # isDerivedFrom
```

Read in prose, an instance `rak:P100001` of a sampling process targets the site `rak:1000001` and yields a collection `rak:7000001` as its output; the deep soil sample `rak:6000001` is in turn asserted to be derived from the same site. The redundancy between `hasTarget` on the process and `isDerivedFrom` on the sample is intentional: the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.

### 4.3.3 Experimental protocol pattern

We model experimental protocols as processes specified by a protocol entity. Each process (for example DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO's general PROCESS–HASINPUT/HASOUTPUT–OBJECT idiom and adds a specification link to a reusable protocol description.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \tag{10}$$
$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \tag{11}$$
$$\text{ProtocolExecution} \sqsubseteq \exists\text{isSpecifiedBy.Protocol} \tag{12}$$
$$\text{ProtocolExecution} \sqsubseteq \exists\text{hasInput.MaterialEntity} \sqcap \exists\text{hasOutput.MaterialEntity} \tag{13}$$

Listing 3 demonstrates the formalization of a DNA extraction process.

Listing 3: Turtle representation of a DNA extraction protocol.

```
rak:P200001 a sio:000994 ; # Scientific Protocol Execution
  sio:000339 rak:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
  sio:000008 rak:4900001 . # hasAttribute (Concentration Quality)
```

In words, the protocol-execution instance `rak:P200001` is specified by the named extraction protocol `rak:DNA_ext_powersoil`, takes the soil sample `rak:6000001` as input, and produces the DNA extract `rak:6800001` as output; the resulting extract carries a concentration quality (`rak:4900001`) which is itself quantified through a separate measurement process following the pattern of Equation 1.

#### 4.3.4 Taxonomic abundance pattern

Taxonomic profiles represent collections of taxon observations containing associated relative and absolute abundances. Instead of modeling the feature table as a flat matrix, we formalize the entire table as a bioinformatic workflow that produces a dataset containing individual taxonomic observations. We use the merged taxonomy (see the companion data paper [12]) to assign specific NCBI or GTDB classes as the target of each observation. For instance, the sequence analysis for a specific sample yields an absolute sequence count associated with a taxonomic class from NCBI Taxonomy. We construct an abundance quality that targets the explicit taxonomic identifier, binding the observational count back to the biological workflow. Formally, the pattern instantiates SIO's Collection–hasMember–Thing and Quantity–refersTo–Quality idioms:

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \tag{14}$$

$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \tag{15}$$

$$\text{BioinformaticWorkflow} \sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset} \tag{16}$$

$$\text{TaxonomicDataset} \sqsubseteq \forall \text{hasMember.AbundanceObservation} \tag{17}$$

$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists \text{refersTo.AbundanceQuality} \tag{18}$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{isAttributeOf.Taxon} \tag{19}$$

Listing 4: Turtle representation of taxonomic abundance formalized from a feature table.

```
rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000230 rak:7900001 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:7290001 . # hasOutput (Taxonomic Dataset)

rak:7290001 a rak:0000074 ; # Taxonomic Dataset
  sio:000059 rak:4400001 . # hasMember (Observation Value)

rak:4400001 a sio:000070 ; # Quantity (Absolute Count)
  sio:000215 rak:5400001 ; # refersTo
  rak:2000021 "15"^^xsd:integer . # absolute read count

rak:5400001 a rak:0000078 ; # Absolute Abundance Quality
  sio:000011 ncbitaxon:286 . # isAttributeOf (Pseudomonas)
```

Verbalized, the bioinformatic-workflow instance `rak:P290001` consumes a FASTQ dataset (`rak:7900001`) and produces the taxonomic dataset `rak:7290001`; that dataset has as one of its members the quantity `rak:4400001`, which records an absolute read count of `15` and refers to an absolute abundance quality that is in turn an attribute of *Pseudomonas* (`ncbitaxon:286`). One row of a feature table is therefore expanded into a four-node sub-graph that preserves both the classifier output and its provenance.

#### 4.3.5 XRF geochemistry pattern

We model XRF analysis as a process that targets a soil sample and produces multiple measurement values, each corresponding to the concentration of a specific chemical analyte. We map each analyte to the Chemical Entities of Biological Interest (ChEBI) ontology [7] and to the PubChem database [26]. Out of 93 monitored geochemical parameters, we mapped 80 analytes directly to corresponding ChEBI classes and aligned 92 parameters to PubChem Compound

Identifiers. The XRF pattern is a specialization of the measurement pattern (Equation 1), with the target restricted to a soil sample and the produced quantity bound to an analyte-specific quality:

$$\text{XRFMeasurementProcess} \sqsubseteq \text{MeasurementProcess} \tag{20}$$

$$\text{XRFMeasurementProcess} \sqsubseteq \exists\text{hasTarget.SoilSample} \sqcap \exists\text{hasOutput.ConcentrationQuantity} \tag{21}$$

$$\text{ConcentrationQuantity} \sqsubseteq \text{Quantity} \sqcap \exists\text{refersTo.AnalyteQuality} \tag{22}$$

$$\text{AnalyteQuality} \sqsubseteq \exists\text{isAttributeOf.SoilSample} \tag{23}$$

Listing 5: Turtle representation of an XRF geochemistry measurement.

```
rak:P300001 a rak:0000080 ; # XRF Measuring Process
  sio:000291 rak:6000001 ; # hasTarget
  sio:000229 rak:4300001 . # hasOutput

rak:4300001 a sio:000070 ; # Quantity
  sio:000215 rak:5300001 ; # refersTo
  sio:000221 uo:0000187 ; # Percent unit
  rak:2000012 "58.2"^^xsd:double . # Concentration

rak:5300001 a rak:0000167 ; # Analyte Quality Class
  sio:000011 rak:6000001 . # isAttributeOf
```

In words, the XRF measurement process `rak:P300001` targets the soil sample `rak:6000001` and produces the quantity `rak:4300001`, which records a concentration of `58.2` percent (`uo:0000187`) and refers to an analyte-specific quality `rak:5300001`. Because the analyte quality is typed with a dedicated subclass per element, the same pattern instance can be reused unchanged across all 91 geochemical analytes by varying only the quality class IRI.

#### 4.3.6 Sampling site and biome assignment pattern

Each physical sampling location is represented as a persistent individual that is grounded in the Environment Ontology (ENVO) by linking to a biome and to one or more environmental features. We use both an existential and a universal restriction so that the asserted biome both holds and is the only admissible value, which preserves the closed-world intent of the field record.

$$\text{SamplingSite} \sqsubseteq \text{Site} \tag{24}$$

$$\text{SamplingSite} \sqsubseteq \exists\text{hasBiome.ENVO:Biome} \sqcap \forall\text{hasBiome.ENVO:Biome} \tag{25}$$

$$\text{SamplingSite} \sqsubseteq \exists\text{hasEnvironmentalFeature.ENVO:EnvironmentalFeature} \tag{26}$$

Listing 6: Turtle representation of a sampling site grounded in ENVO.

```
rak:1000001 a rak:0000002 ; # Sampling Site
  rdfs:label "Site 11" ;
  geo:asWKT "POINT(50.123 22.456)"^^geo:wktLiteral ;
  rak:2000050 envo:01000219 ; # hasBiome (desert biome)
  rak:2000051 envo:00000086 . # hasEnvironmentalFeature (sand dune)
```

In words, the sampling site `rak:1000001` carries a label, a WKT geometry, and is asserted both to belong to the ENVO desert biome and to exhibit a sand-dune environmental feature. The combined existential and universal restriction on `hasBiome` (Equation 24) ensures that any reasoner treats the asserted biome as the only admissible value for that site.

#### 4.3.7 Sequencing workflow pattern

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of PROCESS–HASINPUT/HASOUTPUT–OBJECT steps: a DNA extraction process consumes a soil sample and produces a DNA extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern of Equation 10.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \tag{27}$$
$$\text{DNAExtractionProcess} \sqsubseteq \exists \text{hasInput.SoilSample} \sqcap \exists \text{hasOutput.DNAExtract} \tag{28}$$
$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \tag{29}$$
$$\text{LibraryPreparationProcess} \sqsubseteq \exists \text{hasInput.DNAExtract} \sqcap \exists \text{hasOutput.AmpliconLibrary} \tag{30}$$
$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \tag{31}$$
$$\text{SequencingProcess} \sqsubseteq \exists \text{hasInput.AmpliconLibrary} \sqcap \exists \text{hasOutput.FASTQDataset} \tag{32}$$
$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \sqcap \forall \text{hasMember.SequenceRead} \tag{33}$$

Listing 7: Turtle representation of the sequencing workflow from soil sample to FASTQ dataset.

```
rak:P200001 a rak:0000044 ; # DNA Extraction Process
  sio:000339 rak:L000001 ; # isSpecifiedBy (PowerSoil protocol)
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000045 ; # Library Preparation Process
  sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
  sio:000230 rak:6800001 ; # hasInput (DNA Extract)
  sio:000229 rak:6900001 . # hasOutput (Amplicon Library)

rak:P280001 a rak:0000046 ; # Sequencing Process
  sio:000230 rak:6900001 ; # hasInput (Amplicon Library)
  sio:000229 rak:7900001 . # hasOutput (FASTQ Dataset)

rak:7900001 a rak:0000064 ; # FASTQ Dataset
  rdfs:seeAlso <https://www.ebi.ac.uk/ena/browser/view/ERR1234567> .
```

Read as a chain, the DNA extraction process `rak:P200001` consumes the soil sample `rak:6000001` and produces the DNA extract `rak:6800001`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6900001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7900001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput`/`hasOutput` pattern, so a single SPARQL traversal can recover the full provenance from any node.

#### 4.3.8 Sequencing quality control pattern

Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the generic measurement pattern of Equation 1. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \tag{34}$$

$$\text{SequencingQCProcess} \sqsubseteq \exists\text{hasInput.FASTQDataset} \sqcap \exists\text{hasOutput.QCMeasurementValue} \tag{35}$$

$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists\text{refersTo.QCQuality} \tag{36}$$

$$\text{QCQuality} \sqsubseteq \exists\text{isAttributeOf.FASTQDataset} \tag{37}$$

Listing 8: Turtle representation of a sequencing QC measurement (forward-read GC content).

```
rak:P350001 a rak:0000047 ; # Sequencing QC Process
  sio:000230 rak:7900001 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:4500001 . # hasOutput (QC Value)

rak:4500001 a rak:0000086 ; # GC Content Value
  sio:000215 rak:5500001 ; # refersTo
  sio:000221 uo:0000187 ; # Percent
  rak:2000073 "54.2"^^xsd:double . # has GC content

rak:5500001 a rak:0000074 ; # GC Content Quality (Forward)
  sio:000011 rak:7900001 . # isAttributeOf (FASTQ Dataset)
```

In words, the QC process `rak:P350001` takes the FASTQ dataset `rak:7900001` as input and produces the measurement value `rak:4500001`, which records a GC content of `54.2` percent and refers to a forward-read GC-content quality `rak:5500001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation.

### 4.4 OWL and RDF implementation

We generated the OWL/RDF implementation using custom Groovy scripts built on the OWL API 5 [21]. The knowledge base employs a modular design, separating the base ontology, sites, samples, measurements, XRF data, DNA extraction records, sequencing metadata, quality control metrics, and taxonomic abundances into individual ABox modules [18]. Each module is generated by an independent script, validated against its ShEx shape, and loaded into Virtuoso as a separate transaction, so that a single data-type update can be rolled out without rebuilding the full graph.

We use the OpenLink Virtuoso triple store [35] for data hosting and query answering because it is available as free software under the GNU General Public License, supports OWL-Horst forward-chaining materialization, and has demonstrated operational maturity in large-scale linked-data deployments. Virtuoso features an inference engine that can be configured with a custom rule set; we bind the project ontology graph as a rule set named `rubalkhali` that covers all three named graphs (knowledge base, ontology, taxonomy) and provides backward-chaining expansion for rdfs:subClassOf, rdfs:subPropertyOf, owl:inverseOf, owl:equivalentClass, and related OWL-Horst rules.

Prior to OWL-Horst inference, the base knowledge graph comprised 37,043,525 asserted triples across three named graphs (knowledge base 36,978,835; taxonomy 42,406; ontology 16,643), with taxonomy assertions accounting for approximately 36.5 million of the knowledge base graph. During deployment, the workflow applies OWL-Horst (pD*) forward-chaining materialization in three rounds. Round 1 applies the ground rules that pair a single asserted triple with a TBox axiom: `rdfs:subClassOf`, `rdfs:subPropertyOf`, `rdfs:domain`, `rdfs:range`, `owl:inverseOf`, `owl:SymmetricProperty`, `owl:equivalentClass`, and `owl:equivalentProperty`. Round 2 re-applies `subPropertyOf` and `subClassOf` over the triples produced in round 1 so that subproperty chains propagate through the inferred graph. Round 3 applies one-hop transitive closure for all `owl:TransitiveProperty` instances. Each round is implemented as a single

SPARQL INSERT INTO GRAPH statement wrapped in DEFINE input:inference, and executes with row-level logging temporarily disabled (log_enable(2,1)) so that large transactions do not exceed Virtuoso's per-transaction log size limit.

This process generates 80,907,421 materialized triples for a total of 117,950,946 triples (a 3.18× expansion). Materialization completes in approximately 114 minutes on a server with 1 TB RAM and writes the materialized triples to a dedicated named graph (<https://rubalkhali.science/inferred so that query answering can be served from materialized facts without paying a per-query backward-chaining cost. The dominant inferred predicates are sio:isRelatedTo (22.6M), rdf:type (14.6M), sio:hasAttribute (11.3M), sio:isAttributeOf (5.7M), and sio:hasMember (5.6M), reflecting the propagation of SIO's property hierarchy and inverse properties across the ABox.

### 4.5 Schema definition with ShEx

We used Shape Expressions (ShEx) [37] to define structural schemas for the core entity types. While OWL provides open-world reasoning and inference capabilities, it cannot readily enforce data completeness constraints. We developed ShEx schemas to explicitly validate the graph's structure, ensuring data quality and compliance before loading the data into the database.

Our ShEx implementation defines shapes for sampling sites, soil samples, environmental measurements, DNA extracts, amplicon libraries, sequencing datasets, quality control records, and taxonomy observations. These shapes guarantee that essential attributes are invariably present. For instance, Listing 9 shows the shape for a sampling site, which requires an explicit type assertion and a label while permitting optional coordinates and a free-text description.

Listing 9: ShEx shape for Sampling Site validation.

```
<#SamplingSiteShape> {
    rdf:type [rak:0000002] ;  # Must be of type Sampling Site
    rdf:type . * ;                # May have other inferred types
    rdfs:label . ;                # Must have a label
    dcterms:description xsd:string ? ; # Optional description
    geo:asWKT . ? ;               # Optional geographic coordinates
}
```

In words, <#SamplingSiteShape> requires that every conforming node carry an explicit type assertion to rak:0000002 (Sampling Site) and an rdfs:label; it permits any number of additional inferred types and optionally allows a free-text description and a WKT geometry. Records that omit the mandatory type or label are rejected before they reach the triple store.

We integrate the ShEx validation into the automated deployment workflow so that every ABox module is tested against its shape immediately after generation and before loading into Virtuoso. We implement the validator with the Apache Jena ShEx library (version 4.10.0) invoked from Groovy. The workflow terminates with an explicit error if any instance violates its shape (for example a sampling site lacking GPS coordinates, or a DNA extract lacking a concentration value), which functions as a regression test guaranteeing that modifications to the ETL scripts cannot silently introduce structural errors.

## 5 Evaluation

We evaluated the deployed knowledge graph using four complementary approaches: competency question evaluation via SPARQL queries with known expected answers; structural validation of every RDF module against ShEx shapes; logical consistency checking with the ELK reasoner, run both before and after OWL-Horst materialization; and usability assessment through a live web portal.

### 5.1 Competency questions and SPARQL queries

We defined nine competency questions expressed as SPARQL queries [20] with known expected answers and automated their execution against the deployed endpoint so that they serve both as documentation of the intended query workload and as regression tests during ontology updates. The questions cover retrieval of XRF measurements, listing sampling sites with coordinates, extraction of biome assignments via OWL restrictions, retrieval of Light Elements percentages, DNA extraction records, provenance tracing from soil samples to sequencing runs, taxonomic composition per run, a cross-domain join between environmental temperature and taxonomic abundance, and per-sample sequencing quality control metrics. We present three representative listings below; the full nine-query suite is distributed with the source code [10, 11] and is pre-loaded as click-to-load examples in the web portal's SPARQL editor.

Listing 10 retrieves XRF measurements for a specific site, requiring traversal of the measurement process, quality, and value entities.

Listing 10: SPARQL query to retrieve field XRF measurements for Site 10.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?processLabel ?analyte ?concentration ?unitLabel
WHERE {
  ?sample a rak:0000021 ;
          rdfs:label ?sampleLabel .
  FILTER(REGEX(?sampleLabel, "Site 10"))
  ?value sio:000232 ?process .
  ?process rdfs:label ?processLabel .
  FILTER(REGEX(?processLabel, "Field", "i"))
  ?sample sio:000008 ?quality .
  ?quality sio:000215 ?value .
  ?value rak:2000012 ?concentration .
  ?value sio:000221 ?unit .
  ?unit rdfs:label ?unitLabel .
  ?quality a ?qualityClass .
  ?qualityClass rdfs:label ?analyte .
  FILTER(?qualityClass != rak:0000029)
}
ORDER BY ?processLabel ?analyte LIMIT 100
```

Read in prose, the query first selects every soil sample whose label matches `Site 10`, then walks back from each measurement value to its producing process and forwards from the sample through its quality and value to the recorded concentration and unit. The final filter excludes the Light Elements pseudo-analyte (`rak:0000029`) so that only individually quantified elements appear in the result.

Listing 11 retrieves taxonomic abundance profiles, joining across bioinformatic workflows, FASTQ datasets, and taxon observations.

Listing 11: SPARQL query to retrieve taxonomic composition and abundance for all samples.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT DISTINCT ?run ?taxonLabel ?count ?relativeAbundance
WHERE {
  ?proc a rak:0000071 ;
        sio:000230 ?fastq ;
        sio:000229 ?dsAbs , ?dsRel .
  ?dsAbs a rak:0000074 .
  ?dsRel a rak:0000075 .
  ?fastq rdfs:label ?runLabel .
  BIND(REPLACE(?runLabel, "FASTQ dataset for ", "") AS ?run)
  ?dsAbs sio:000059 ?valAbs .
  ?valAbs rak:2000021 ?count .
  ?valAbs sio:000215 ?qualAbs .
  ?qualAbs sio:000011 ?taxon .
  ?taxon rdfs:label ?taxonLabel .
```

```
  ?qualRel sio:000011 ?taxon ;
           a rak:0000072 ;
           sio:000214 ?valRel .
  ?valRel rak:2000020 ?relativeAbundance .
  FILTER(?taxon != ?fastq)
}
ORDER BY ?run DESC(?count) LIMIT 100
```

The query identifies every bioinformatic workflow that produces both an absolute and a relative taxonomic dataset for the same FASTQ input, then retrieves, for each absolute observation, the read count, the associated taxon label, and the matching relative abundance from the parallel relative dataset. Results are grouped by sequencing run and ordered by descending count.

Listing 12 shows a cross-domain query that joins environmental temperature with taxonomic abundance—a join that would require at least four separate tables in a relational database.

Listing 12: SPARQL query joining environmental temperature and *Pseudomonas* relative abundance.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX pato: <http://purl.obolibrary.org/obo/PATO_>
PREFIX ncbitaxon: <http://purl.obolibrary.org/obo/NCBITaxon_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?siteLabel ?temp ?relAbundance
WHERE {
  ?site a rak:0000002 ;
        rdfs:label ?siteLabel .
  ?tempProcess sio:000291 ?site ;
               sio:000229 ?tempVal .
  ?tempVal rak:2000012 ?temp ;
           sio:000215 ?tempQual .
  ?tempQual a pato:0000146 .
  FILTER(?temp > 35.0)
  ?sample sio:000252 ?site .
  ?ext sio:000230 ?sample ; a rak:0000041 ; sio:000229 ?dna .
  ?libPrep sio:000230 ?dna ; a rak:0000062 ; sio:000229 ?lib .
  ?seq sio:000230 ?lib ; a rak:0000064 ; sio:000229 ?fastq .
  ?bioinf a rak:0000071 ;
          sio:000230 ?fastq ;
          sio:000229 ?dsRel .
  ?dsRel a rak:0000075 ;
         sio:000059 ?valRel .
  ?valRel rak:2000020 ?relAbundance ;
          sio:000215 ?qualRel .
  ?qualRel sio:000011 ncbitaxon:286 . # Pseudomonas
}
```

The query first selects every sampling site at which the recorded environmental temperature exceeds 35 °C, then for each such site it walks the full provenance chain—soil sample, DNA extract, library preparation, sequencing run, FASTQ dataset, bioinformatic workflow, relative abundance dataset—down to the relative abundance value of *Pseudomonas* (`ncbitaxon:286`). Each row of the result therefore reports a single hot site together with its corresponding *Pseudomonas* relative abundance.

### 5.2 ShEx validation results

We validated the structural integrity of every RDF module against its ShEx shape as described in Section 4.5. All eight shape-annotated modules (`rubalkhali_sites.owl`, `rubalkhali_samples.owl`, `rubalkhali_measurements.owl`, `rubalkhali_dna.owl`, `rubalkhali_xrf.owl`, `rubalkhali_taxonomy_abox.ow`
`rubalkhali_sra.owl`, and `rubalkhali_qc.owl`) passed validation without constraint violations. Because the validation step is part of the deployment workflow and the workflow aborts on any violation, any subsequent modification to the ETL scripts that introduces a structural error is caught before the data reaches Virtuoso.

## 5.3 Logical consistency with ELK

We verified the logical consistency of the knowledge graph using the ELK reasoner [24], an implementation of a consequence-based reasoning procedure for the OWL 2 EL profile. We loaded the complete set of OWL modules (base ontology, all ABox files, and the taxonomy reference graph) into a single OWL ontology and ran ELK to check for unsatisfiable classes and contradictory axioms. We ran the reasoner in two configurations: (1) on the asserted knowledge graph prior to materialization, and (2) on the union of the asserted knowledge graph and the OWL-Horst materialized graph from Section 4.4. Table 3 reports the results.

In both cases the reasoner confirmed that the ontology is consistent and reported zero unsatisfiable classes. The post-materialization run loaded 81.3 million axioms (of which 78.1 million originate from the materialized ABox), discovered 74 additional classes (7,202 $\rightarrow$ 7,276) from the materialized type assertions, and required 63 seconds for the consistency check. These results confirm that the OWL-Horst entailments introduced no disjointness violations despite adding 5.7 million new individual type assertions, a property that cannot be guaranteed by inspection alone: new ABox type triples of the form $C(a)$ and $D(a)$ where $C \sqsubseteq \neg D$ would render the ontology inconsistent, so the reasoner must be run after materialization to rule this out.

Table 3: ELK reasoning performance before and after OWL-Horst materialization. All runs used OWL API 4.5.26 and ELK 0.4.3 on a server with 1 TB RAM (Java heap 850 GB).

| Metric | Before | After |
|---|---|---|
| Modules loaded | 8 | 10 |
| Total axioms | 224,775 | 81,273,368 |
| Logical axioms | 124,106 | 17,693,762 |
| Classes | 7,202 | 7,276 |
| Named individuals | 30,680 | 5,712,603 |
| `isConsistent` | 0.84 s | 63.4 s |
| Class hierarchy | 0.40 s | 0.73 s |
| Unsatisfiable classes | 0 | 0 |
| Load time | 675 s | 7,585 s |
| Peak RSS | 203 GB | 395 GB |
| Wall time | 688 s | 7,722 s |

We chose ELK over more expressive reasoners such as HermiT [17] because the combination of TBox and 37-million-triple ABox renders full OWL 2 DL reasoning computationally infeasible on commodity hardware. The consequence-based calculus implemented by ELK produces class-hierarchy results in under two seconds on the asserted graph, and the post-materialization run confirms the same consistency result at the cost of a higher load time and increased memory.

## 5.4 Compute-time metrics

Table 4 summarizes the computational cost of the key deployment steps. OWL-Horst forward-chaining materialization runs in approximately 114 minutes and expands the knowledge graph from 37.0 million asserted triples to 117.9 million (3.18×). ELK consistency checking over the asserted TBox takes under two seconds; the post-materialization run that includes the full materialized ABox requires 63 seconds but confirms the same result (zero unsatisfiable classes). Competency question latency on the Virtuoso endpoint ranges from 5 ms for simple site lookups (CQ2) to over one second for queries that traverse inverse properties across the entire DNA extraction workflow (CQ5).

The taxonomy composition query (CQ7) exceeds a 5-minute timeout because it joins absolute and relative abundance datasets across 1.4 million observations and therefore touches a large fraction of the materialized graph. In practice this query is not used by the web portal; the

Table 4: Compute-time metrics for key deployment and validation steps.

| **Step** | **Time** |
|---|---|
| RDF module loading (Virtuoso, 9 ABox files) | 140 s |
| OWL-Horst materialization (3 rounds) | 114 min |
| ELK classification (asserted only) | 1.6 s |
| ELK classification (with materialized ABox) | 64 s |
| CQ1 – XRF measurements, Site 10 | 23 ms |
| CQ2 – Sampling sites + GPS | 5 ms |
| CQ3 – Biomes per site | 5 ms |
| CQ4 – Light Elements percentages | 8 ms |
| CQ5 – DNA extraction records | 1,342 ms |
| CQ6 – Provenance chain | 255 ms |
| CQ7 – Taxonomic composition | >300 s |

portal retrieves taxonomic data per sample rather than across all runs, which completes in sub-second time. CQ7 is retained as a stress test for the deployment and as a marker of the frontier of what can be served interactively from the current hardware.

### 5.5 Web portal

We developed a web portal (`https://rubalkhali.science/`) that provides user-friendly access to the knowledge graph. The portal is implemented as a React front-end backed by a FastAPI service and Virtuoso: the backend dynamically generates SPARQL queries to fetch content directly from the knowledge graph in real time, without maintaining a separate relational database. Every page rendered by the portal constitutes a functional test of the underlying data, so that any data integrity problem surfaces as a visible error. The portal provides an interactive map of the 70 sampling sites, per-site dashboards with environmental measurements, geochemistry, and taxonomic profiles, and a SPARQL query page built on YASGUI [39] that provides an autocomplete-aware editor, prefix management, and a tabbed interface. The nine competency questions of Section 5 are exposed as click-to-load examples in the SPARQL page so that users can run them against the live endpoint without typing. The portal additionally offers a downloads page that serves the individual OWL/RDF modules, the ShEx shape definitions, and the competency questions as static files under the same free software licence as the rest of the resource.

## 6 Discussion

We revisit the challenges introduced in the introduction, link each to the design choices described in the preceding sections, and conclude with a discussion of limitations and sustainability.

### 6.1 Design rationale

**Unified data model.** The first challenge was to represent heterogeneous observation types under a single vocabulary. We adopted SIO as the upper-level framework precisely because it offers a small set of relations (`hasInput`/`hasOutput`, `hasTarget`, `hasMember`, `refersTo`, `hasValue`) that generalize across sampling, measurement, and provenance patterns. Sample collection, XRF analysis, DNA extraction, sequencing, and taxonomic classification are all instantiations of the same Process–hasInput/hasOutput–Object idiom; quantitative results are uniformly represented as Quantity–refersTo–Quality pairs with explicit units; and feature tables are represented as Collection–hasMember–Thing structures. This uniformity is what allows cross-domain SPARQL queries to traverse from a climate observation through a sampling site to a taxon abundance without any per-domain join logic, as demonstrated by the query in Listing 12. Recent work on simplified upper-level ontologies, notably SULO [9], takes this argument further: by strengthening a few fundamental relations and reducing reliance on specialized predicates, an upper-level ontology can lower the modelling effort for new domains while preserving interoperability. Our patterns are compatible with this direction and could be re-grounded in SULO with limited refactoring.

**Taxonomy harmonization.** The companion data paper [12] describes our mapping of operational taxonomic units from multiple primer/classifier combinations onto NCBI Taxonomy and GTDB. A key design choice is that we keep both the raw classifier output and the harmonized identifier in the knowledge graph, so that downstream consumers can re-use the mapping or substitute their own without losing the original assertions. This is important for reproducibility: because classifier outputs are sensitive to reference database version, a consumer that reruns the classification against a newer reference may legitimately produce a different taxon assignment for the same read, and the knowledge graph must allow both assignments to coexist.

**Schema validation at ingest.** We chose ShEx for ingest validation because the ETL is batch-oriented and the validator runs in the same Groovy/Jena toolchain that produces the RDF. SHACL is an obvious alternative and is supported in-database by several triple stores; a future port of the shapes from ShEx to SHACL would enable in-database validation at load time and is a natural extension of the current workflow. We deployed Virtuoso as the triple store because it is available as free software under the GNU General Public License, supports OWL-Horst forward-chaining materialization, and has demonstrated operational maturity in large-scale linked-data deployments.

**Semantic query answering at scale.** Full OWL 2 DL reasoning over the combined TBox and 37-million-triple ABox is not computationally feasible with HermiT on commodity hardware. We therefore split the reasoning task. We use ELK [24] to *validate* the logical consistency of the knowledge graph (confirming zero unsatisfiable classes both before and after materialization); we use Virtuoso's OWL-Horst implementation to *materialize* entailments under pD* semantics so that query answering can be served from physical triples without a per-query backward-chaining cost. This split is deliberate: ELK is sound and complete for classification in OWL 2 EL and therefore gives us a trustworthy "no contradiction" result, but its consequence-based calculus does not directly produce the kind of ABox expansion (inverse properties, subproperty chains) that SPARQL traversal queries benefit from. OWL-Horst, conversely, covers the rules that matter for query answering—subclass, subproperty, domain, range, inverse, symmetry, equivalence, transitivity—but is incomplete with respect to full OWL 2 DL. Applying both reasoners gives us the best of both: a logically consistent graph whose materialized form exposes exactly the entailments that the competency questions need. Two decades of work on scaling OWL reasoning [24, 32, 43] provide additional alternative points in the design space, and a more expressive reasoner could be applied to a sub-module of the TBox in future work.

**User-facing interface.** The web portal addresses the challenge that domain scientists are typically not fluent in SPARQL. The interactive map and per-site dashboards cover the most common exploratory tasks; the YASGUI-based SPARQL editor exposes the underlying graph for power users, pre-populated with the competency questions as click-to-load examples, autocomplete for prefixes and predicates, and a tabbed query interface so that multiple queries can be developed in parallel.

### 6.2 Limitations and outstanding challenges

Several challenges remain open. First, although OWL-Horst materialization plus ELK classification is sufficient for the queries described in this paper, more complex reasoning—for example over disjunctions arising from ENVO biome classifications, or over negative axioms about taxon absence—would require either a different reasoner or the extraction of a sub-module amenable to a more expressive calculus such as HermiT. Second, the resource depends on upstream ontologies (SIO, ChEBI, ENVO, NCBI Taxonomy) that evolve independently; keeping the deployed graph aligned with their releases is an ongoing maintenance task and a known reproducibility risk for any SPARQL query that bottoms out in a specific external IRI. Third, the 11 GB materialized ABox dump exceeds the in-memory parsing capacity of the OWL API Turtle parser on a workstation with less than 128 GB of RAM, so the post-materialization ELK run currently requires a large-memory server; splitting the materialized graph into stream-parseable chunks would relax this requirement. Fourth, CQ7 (the taxonomy composition query that joins all 1.4 million absolute and relative abundance observations) still exceeds a five-minute timeout on commodity hardware; reducing its cost either by query rewriting or by additional materialization indexing remains future work.

### 6.3 Sustainability and updates

The dataset is versioned and archived on Zenodo and GigaDB, and the deployment workflow is reproducible from the public repository. We intend to extend the resource as additional field expeditions are processed: new trips will be added as additional ABox modules without changes to the upper-level patterns. Upstream ontology updates will be tracked at scheduled intervals; re-materialization is scripted as the three SPARQL Update statements of Section 4.4 and runs as part of the deployment workflow, so an upstream release can be absorbed by re-running the workflow. Where an upstream change would alter the semantics of an existing competency question, we will record the affected query in the changelog so that downstream users can audit the impact.

## 7 FAIR compliance

The design decisions described in the preceding sections collectively ensure that the resource satisfies the FAIR Data Principles [45].

- **Findable.** We assigned globally unique IRIs under the `https://rubalkhali.science/kb/` namespace to all named individuals, and we registered the dataset with ENA under accession PRJEB104209.
- **Accessible.** We serve the knowledge graph via a standard SPARQL 1.1 endpoint and a web interface; raw sequencing reads are deposited at ENA. Every file is dereferenceable and every site and sample IRI resolves to a content-negotiated representation.
- **Interoperable.** We employ established shared vocabularies (SIO, ENVO, ChEBI, PATO, UO, NCBI Taxonomy) and standard formats (OWL 2 DL, RDF/Turtle, SPARQL), and every identifier from external ontologies is used as-is so that federated queries with other SIO- or OBO-based graphs are straightforward.
- **Reusable.** We licensed the data under CC BY 4.0 and described it with rich provenance metadata encoding the complete chain from physical sample to digital data product. We open-sourced all generation code.

## 8 Conclusion

We have presented a formal knowledge graph for the Rub' al-Khali desert ecosystem that integrates microbial sequencing, soil geochemistry, environmental observations, and climate records under a single SIO-based data model. We formalized eight reusable ontology design patterns—measurement, sampling, experimental protocol, taxonomic abundance, XRF geochemistry, sampling site with biome, sequencing workflow, and sequencing quality control—in description logic and aligned them explicitly with SIO's core relational idioms, accompanied by Turtle listings and prose verbalizations for every pattern. We implemented an automated deployment workflow that generates the modular OWL/RDF files, validates them against ShEx shapes as a regression test, loads them into Virtuoso, binds the OWL-Horst rule set, and forward-chains materialization via three SPARQL Update rounds. The materialization expands the graph from 37.0 million asserted to 117.9 million total triples (3.18×) in approximately 114 minutes, and the resulting graph is served through a public SPARQL endpoint and a web portal with an autocomplete-aware SPARQL editor. We validated the logical consistency of the full graph with the ELK reasoner both before and after materialization, confirming zero unsatisfiable classes in both cases. We evaluated nine competency questions covering per-sample retrievals, provenance tracing, and cross-domain joins, with latencies that range from 5 ms for simple site lookups to just over one second for queries that traverse the DNA extraction workflow.

The resource contributes a reusable blueprint for integrating a multi-expedition ecological dataset into a Semantic Web deployment. The ontology design patterns are directly applicable to other sampling-based environmental studies; the OWL-Horst materialization workflow is a concrete recipe for any Virtuoso-based deployment that needs to serve inferred triples without a per-query reasoning cost; and the combination of ELK for consistency validation and OWL-Horst for query-time expansion is a practical template for scaling semantic query answering to tens of millions of triples on commodity hardware. The underlying dataset is described in a companion data paper [12], and the deployed knowledge graph is available at `https://rubalkhali.science/`.

## 9 Availability of source code and requirements

All source code for data processing, ontology engineering, RDF generation, ShEx validation, the SPARQL competency questions, the React+FastAPI web portal, and the Docker Compose deployment is publicly available at `https://github.com/bio-ontology-research-group/empty-quarter` under the MIT licence. The ontology modules, ShEx shapes, competency questions, and full knowledge graph dump are archived on Zenodo [10] and GigaDB [11] under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The public SPARQL endpoint is available at `https://rubalkhali.science/sparql`; the interactive web portal is at `https://rubalkhali.science/`.

## Declarations

**Competing interests.** The authors declare no competing interests.

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## References

[1] Franz Baader, Diego Calvanese, Deborah L McGuinness, Daniele Nardi, and Peter F Patel-Schneider. *The Description Logic Handbook: theory, implementation, and applications.* Cambridge University Press, 2nd edition, 2007.

[2] Anita Bandrowski, Ryan Brinkman, Mathias Brochhausen, Matthew H. Brush, Bill Bug, Marcus C. Chibucos, Kevin Clancy, Mélanie Courtot, Michel Dumontier, Melissa A. Haendel, et al. The Ontology for Biomedical Investigations. *PLOS ONE*, 11(4):e0154556, 2016.

[3] Peter Bauer, Björn Stevens, and Wilco Hazeleger. Destination Earth: a digital twin of Earth for the green transition. *Nature Climate Change*, 11(2):80–83, 2021.

[4] Pier Luigi Buttigieg, Norman Morrison, Barry Smith, Christopher J Mungall, and Suzanna E Lewis. The environment ontology: contextualising biological and biomedical entities. *Journal of Biomedical Semantics*, 4(1):43, 2013.

[5] Benjamin J Callahan, Paul J McMurdie, Michael J Rosen, Andrew W Han, Amy Jo A Johnson, and Susan P Holmes. DADA2: high-resolution sample inference from Illumina amplicon data. *Nature Methods*, 13(7):581–583, 2016.

[6] Werner Ceusters and Barry Smith. Information Artifact Ontology (IAO), 2015. OBO Foundry. `http://www.obofoundry.org/ontology/iao.html`.

[7] Kirill Degtyarenko, Paula de Matos, Marcus Ennis, Janna Hastings, Martin Zbinden, Alan McNaught, Rafael Alcántara, Michael Darsow, Mickaël Guedj, and Michael Ashburner. ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research*, 36(suppl_1):D453–D462, 2008.

[8] Michel Dumontier, Christopher JO Baker, Joachim Baran, Alison Callahan, Leonid Chepelev, José Cruz-Toledo, Nicholas R Del Rio, Geraint Duck, Laura I Furlong, Nichealla Keast, et al. The Semanticscience Integrated Ontology (SIO) for biomedical research and knowledge discovery. *Journal of Biomedical Semantics*, 5(1):14, 2014.

[9] Michel Dumontier, Remzi Çelebi, Komal Gilani, Isabelle de Zegher, Katerina Serafimova, Catalina Martínez Costa, and Stefan Schulz. SULO – a simplified upper-level ontology. In *Proceedings of the Joint Ontology Workshops (JOWO) – Episode XI: The Sicilian Summer under the Etna, co-located with the 15th International Conference on Formal Ontology in Information Systems (FOIS 2025)*, Catania, Italy, September 2025. `https://github.com/AIDAVA-DEV/sulo/blob/main/docs/papers/FOUST2025/SULO_FOUST2025.pdf`.

[10] Empty Quarter Consortium. Empty quarter knowledge graph: ontology, abox modules, shex shapes, and competency questions. Zenodo, 2026. DOI to be assigned upon publication.

[11] Empty Quarter Consortium. Empty quarter: Rub' al-khali desert microbiome and geochemistry dataset. GigaDB, 2026. DOI to be assigned upon publication.

[12] Empty Quarter Consortium. A multi-expedition soil microbiome and geochemistry dataset from the Rub' al-Khali desert, 2026. GigaScience Data Note, companion paper. In preparation.

[13] Philip A Ewels, Alexander Peltzer, Sven Fillinger, Harshil Patel, Johannes Alneberg, Andreas Wilm, Maxime Ulysse Garcia, Paolo Di Tommaso, and Sven Nahnsen. The nf-core framework for community-curated bioinformatics pipelines. *Nature Biotechnology*, 38(3):276–278, 2020.

[14] Noah Fierer. Embracing the unknown: disentangling the complexities of the soil microbiome. *Nature Reviews Microbiology*, 15(10):579–590, 2017.

[15] Georgios V Gkoutos, Eric CJ Green, Ann-Marie Mallon, John M Hancock, and Duncan Davidson. The Phenotype And Trait Ontology (PATO): a standard for cross-species phenotype data integration. *Mammalian Genome*, 16:866–879, 2005.

[16] Georgios V Gkoutos, Paul N Schofield, and Robert Hoehndorf. The Units Ontology: a tool for integrating units of measurement in science. *Database*, 2012:bas033, 2012.

[17] Birte Glimm, Ian Horrocks, Boris Motik, Giorgos Stoilos, and Zhe Wang. HermiT: an OWL 2 reasoner. *Journal of Automated Reasoning*, 53(3):245–269, 2014.

[18] Bernardo Cuenca Grau, Ian Horrocks, Yevgeny Kazakov, and Ulrike Sattler. A survey on ontology modularization. *Theoretical Computer Science*, 391(3):198–233, 2008.

[19] Armin Haller, Krzysztof Janowicz, Simon J.D. Cox, Maxime Lefrançois, Kerry Taylor, Danh Le Phuoc, Joshua Lieberman, Raúl García-Castro, Rob Atkinson, and Claus Stadler. Semantic Sensor Network Ontology, 2017. `https://www.w3.org/TR/vocab-ssn/`.

[20] Steve Harris, Andy Seaborne, and Eric Prud'hommeaux. SPARQL 1.1 Query Language. *W3C Recommendation*, 2013. `https://www.w3.org/TR/sparql11-query/`.

[21] Matthew Horridge and Sean Bechhofer. The OWL API: a Java API for OWL ontologies. *Semantic Web*, 2(1):11–21, 2011.

[22] Félix Iglesias, Frédéric Ros, Lynh Hoang Vy Thuy, Laurence Gourcy, Jean-Sébastien Moquet, Véronique Daële, and Sébastien Dupraz. A conceptual architecture for AI-assisted digital twins in natural resource management. *Ecological Informatics*, 94:103635, 2026.

[23] David Jones, Chris Snider, Aydin Nassehi, Jason Yon, and Ben Hicks. Digital twins: State of the art theory and practice, challenges, and open research questions. *Journal of Manufacturing Systems*, 60:411–432, 2021.

[24] Yevgeny Kazakov, Markus Krötzsch, and František Simančík. The even more irresistible SROIQ. *Proceedings of the Thirteenth International Conference on the Principles of Knowledge Representation and Reasoning*, pages 57–67, 2012.

[25] Taimur Khan, Koen de Koning, Dag Endresen, Desalegn Chala, and Erik Kusch. TwinEco: a unified framework for dynamic data-driven digital twins in ecology. *Ecological Informatics*, 91:103407, 2025.

[26] Sunghwan Kim, Jie Chen, Tiejun Cheng, Asta Gindulyte, Jia He, Siqian He, Qingliang Li, Benjamin A Shoemaker, Paul A Thiessen, Bo Yu, et al. PubChem 2023 update. *Nucleic Acids Research*, 51(D1):D1373–D1380, 2023.

[27] Timothy Lebo, Satya Sahoo, Deborah McGuinness, Khalid Belhajjame, James Cheney, David Corsar, Daniel Garijo, Stian Soiland-Reyes, Stephan Zednik, and Jun Zhao. PROV-O: The PROV ontology, 2013. `https://www.w3.org/TR/prov-o/`.

[28] Joshua Madin, Shawn Bowers, Mark Schildhauer, Sergeui Krivov, Deana Pennington, and Ferdinando Villa. An ontology for describing and synthesizing ecological observation data. *Ecological Informatics*, 2(3):279–296, 2007.

[29] Thulani P Makhalanyane, Angel Valverde, Eoin Gunnigle, Aline Frossard, Jean-Baptiste Ramond, and Don A Cowan. Microbial ecology of hot desert edaphic systems. *FEMS Microbiology Reviews*, 39(2):203–221, 2015.

[30] James P. Mandaville. Plant life in the rub' al-khali (the empty quarter), south-central arabia. *Proceedings of the Royal Society of Edinburgh. Section B. Biological Sciences*, 89:147–157, 1986.

[31] Amy Marshall-Colon, Stephen P Long, Douglas K Allen, Geraldine Allen, Daniel A Beard, Bedrich Benes, Susanne von Caemmerer, A J Christensen, Douglas J Cox, John C Hart, et al. Towards a digital twin for plant biology. *Trends in Plant Science*, 22(12):1024–1033, 2017.

[32] Boris Motik, Bernardo Cuenca Grau, Ian Horrocks, Zhe Wu, Achille Fokoue, and Carsten Lutz. OWL 2 Web Ontology Language Profiles (Second Edition). W3c recommendation, W3C, December 2012. `https://www.w3.org/TR/owl2-profiles/`.

[33] Boris Motik, Bernardo Cuenca Grau, Ian Horrocks, Zhe Wu, Achille Fokoue, and Carsten Lutz. OWL 2 Web Ontology Language document overview. *W3C Recommendation*, 2012. `https://www.w3.org/TR/owl2-overview/`.

[34] Open-Meteo. Open-Meteo: free weather API. `https://open-meteo.com/`, 2024.

[35] OpenLink Software. OpenLink Virtuoso Universal Server. `https://virtuoso.openlinksw.com/`, 2024.

[36] Donovan H Parks, Maria Chuvochina, Christian Rinke, Aaron J Mussig, Pierre-Alain Chaumeil, and Philip Hugenholtz. GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy. *Nucleic Acids Research*, 50(D1):D199–D207, 2022.

[37] Eric Prud'hommeaux, Jose Emilio Labra Gayo, and Harold Solbrig. Shape Expressions: an RDF validation and transformation language. *Proceedings of the 10th International Conference on Semantic Systems*, pages 32–40, 2014.

[38] QUDT Project. QUDT: Quantities, units, dimensions and data types schemas, 2011. `http://www.qudt.org/`.

[39] Laurens Rietveld and Rinke Hoekstra. YASGUI: Feeling the pulse of linked data. In *Knowledge Engineering and Knowledge Management (EKAW 2014)*, volume 8876 of *Lecture Notes in Computer Science*, pages 441–452. Springer, 2014.

[40] Hajo Rijgersberg, Mark van Assem, and Jan Top. Ontology of units of measure and related concepts. *Semantic Web*, 4(1):3–13, 2013.

[41] Conrad L Schoch, Stacy Ciufo, Mikhail Domrachev, Carol L Hotton, Sivakumar Kannan, Oksana Khovanskaya, Detlef Leipe, Richard Mcveigh, Kathleen O'Neill, Barbara Robbertse, et al. NCBI Taxonomy: a comprehensive update on curation, resources and tools. *Database*, 2020:baaa062, 2020.

[42] Daniel Straub, Nia Blackwell, Adrian Langarica-Fuentes, Alexander Peltzer, Sven Nahnsen, and Sara Kleindienst. Interpretations of environmental microbial community studies are biased by the selected 16s rrna (gene) amplicon sequencing pipeline. *Frontiers in Microbiology*, 11:550420, 2020.

[43] Herman J ter Horst. Completeness, decidability and complexity of entailment for RDF Schema and a semantic extension involving the OWL vocabulary. *Journal of Web Semantics*, 3(2–3):79–115, 2005.

[44] John Wieczorek, David Bloom, Robert Guralnick, Stan Blum, Markus Döring, Renato Giovanni, Tim Robertson, and David Vieglais. Darwin Core: An evolving community-developed biodiversity data standard. *PLoS ONE*, 7(1):e29715, 2012.

[45] Mark D Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E Bourne, et al. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3(1):160018, 2016.

[46] Pelin Yilmaz, Renzo Kottmann, Dawn Field, Rob Knight, James R Cole, Linda Amaral-Zettler, Jack A Gilbert, et al. Minimum information about a marker gene sequence (MIMARKS) and minimum information about any (x) sequence (MIxS) specifications. *Nature Biotechnology*, 29(5):415–420, 2011.